

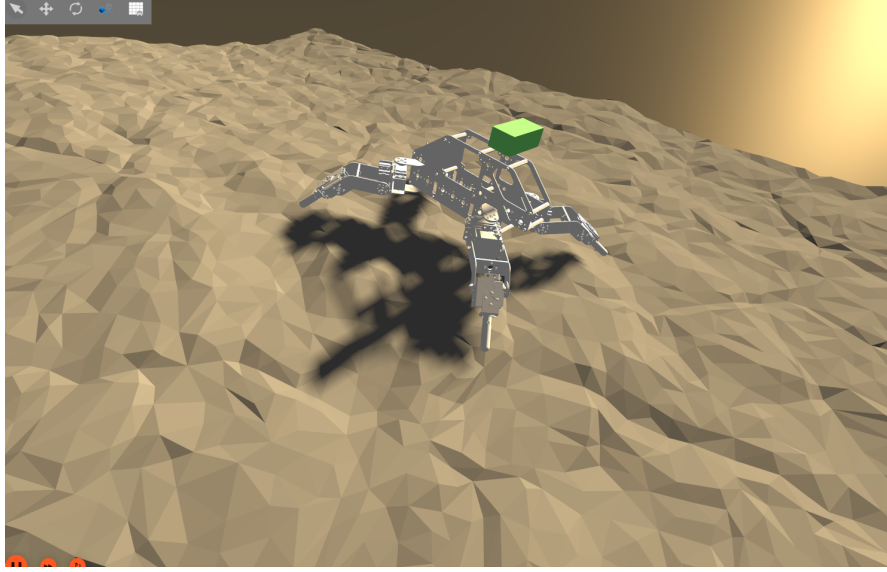


Figure 3.14: Generated rough floor terrain mesh.

Sensors

For joint state feedback, we directly used the joint states published by Gazebo. For IMU data, we used Gazebo’s built-in IMU sensor plugin, which gives readings for linear acceleration, angular velocity, and orientation.

To simulate a depth camera, we used Gazebo’s built-in depth camera plugin. We adjusted the camera’s properties to match those of the ZED2i stereo camera.

Docker

We also set up a Docker container for the simulation environment to allow for easy setup and consistency across different machines. The container includes all the necessary dependencies for running the Gazebo simulation, as well as ROS2 Humble and the necessary packages for running any algorithms.

3.1.3 Algorithms

Open-loop kinematic gait

Below we describe the derivation of the forward and inverse kinematic models of the T-Quad robot. Figure 3.15 shows the top view of the robot, which has the body frame attached at the center of mass (CoM). The legs are labeled as FR (front right), BR

(back right), BL (back left), and FL (front left) with numbers 0-3, respectively, starting clockwise from the top right leg. The robot motion is intended to be forward motion in the y-direction.

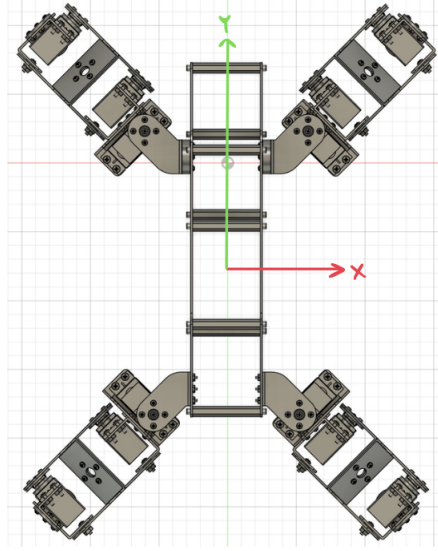


Figure 3.15: Top view of robot body

All joints are modeled as revolute joints. We have assigned 5 frames to each leg, which will give us 4 transformation matrices. Standard DH convention is used to construct the kinematic model.

The base frame of a leg is placed on the hip joint (frame 0, first servo). Subsequent frames are placed on the knee (frame 1, second servo), bend (frame 2, the bend on the physical link at 45°), the ankle (frame 3, third servo), and the tip of the foot (frame 4). The **frame arrangement for the right leg** is shown in Figure 3.16.

By using standard DH convention, we can determine the DH parameters for the right leg as shown in Table 3.1.

r	α	d	θ
l_1	$\pi/2$	0	θ_1
l_2	π	0	θ_2
l_3	0	0	θ_3
l_4	0	0	θ_4

Table 3.1: DH parameters for the right leg.

We can substitute the DH parameters from the table in the transformation matrix for DH parameters:

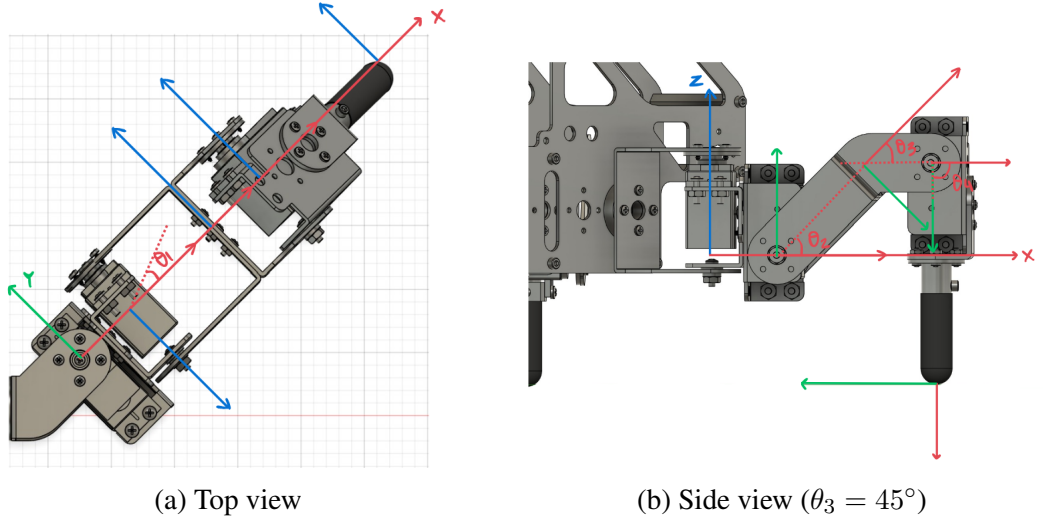


Figure 3.16: Views of the right leg

$${}^{n-1}T_n = \left(\begin{array}{ccc|c} \cos \theta_n & -\sin \theta_n \cos \alpha_n & \sin \theta_n \sin \alpha_n & r_n \cos \theta_n \\ \sin \theta_n & \cos \theta_n \cos \alpha_n & -\cos \theta_n \sin \alpha_n & r_n \sin \theta_n \\ 0 & \sin \alpha_n & \cos \alpha_n & d_n \\ \hline 0 & 0 & 0 & 1 \end{array} \right)$$

to obtain transformation matrices ${}^0T_1, {}^1T_2, {}^2T_3, {}^3T_4$ as follows:

$${}^0T_1 = \left(\begin{array}{ccc|c} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right)$$

$${}^1T_2 = \left(\begin{array}{ccc|c} \cos \theta_2 & 0 & \sin \theta_2 & l_1 \cos \theta_2 \\ \sin \theta_2 & 0 & -\cos \theta_2 & l_1 \sin \theta_2 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right)$$

$${}^2T_3 = \left(\begin{array}{ccc|c} \cos \theta_3 & -\sin \theta_3 & 0 & l_2 \cos \theta_3 \\ \sin \theta_3 & \cos \theta_3 & 0 & l_2 \sin \theta_3 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right)$$

$${}^3T_4 = \left(\begin{array}{ccc|c} 1 & 0 & 0 & l_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right)$$

From these matrices, we can construct the **forward kinematic model** of the right legs (FR & BR) as:

$${}^0T_4 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4$$

$${}^0T_4 = \left(\begin{array}{cccc} \cos(\theta_1 + \theta_2) \cos(\theta_3) & -\cos(\theta_1 + \theta_2) \sin(\theta_3) & \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \cdot C \\ \sin(\theta_1 + \theta_2) \cos(\theta_3) & -\sin(\theta_1 + \theta_2) \sin(\theta_3) & -\cos(\theta_1 + \theta_2) & \sin(\theta_1 + \theta_2) \cdot C \\ \sin(\theta_3) & \cos(\theta_3) & 0 & \sin(\theta_3)(l_2 + l_3) \\ 0 & 0 & 0 & 1 \end{array} \right)$$

where

$$C = l_1 + l_2 \cos(\theta_3) + l_3 \cos(\theta_3)$$

Thus, the position of the foot, given the joint angles, can be extracted from the last column of the transformation matrix:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos(\theta_1 + \theta_2) \cdot l_1 + l_2 \cos(\theta_3) + l_3 \cos(\theta_3) \\ \sin(\theta_1 + \theta_2) \cdot l_1 + l_2 \cos(\theta_3) + l_3 \cos(\theta_3) \\ \sin(\theta_3)(l_2 + l_3) \end{pmatrix}$$

Now, for constructing the **inverse kinematic model**, we need to solve for the joint

angles given the desired foot position. For this, we pre-multiply both sides with the inverse 1T_0 :

$${}^1T_0 {}^0T_4 = {}^1T_0 {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4$$

$${}^1T_0 {}^0T_4 = {}^1T_2 {}^2T_3 {}^3T_4$$

Solving this in MATLAB, we get a set of equations, from which we only extract the fourth columns to get the position of the foot in terms of the joint angles:

$$\begin{pmatrix} x \cos(\theta_1) - l_1 + y \sin(\theta_1) \\ z \\ x \sin(\theta_1) - y \cos(\theta_1) \end{pmatrix} = \begin{pmatrix} l_4 \cos(\theta_3 - \theta_2 + \theta_4) + l_2 \cos(\theta_2) + l_3 \cos(\theta_2 - \theta_3) \\ l_2 \sin(\theta_2) - l_4 \sin(\theta_3 - \theta_2 + \theta_4) + l_3 \sin(\theta_2 - \theta_3) \\ 0 \end{pmatrix} \quad (3.1)$$

To solve for θ_1 , we pick the last equation from 3.1:

$$x \sin \theta_1 - y \cos \theta_1 = 0$$

$$\boxed{\theta_1 = \tan^{-1} \left(\frac{y}{x} \right)} \quad (3.2)$$

Solving for θ_4 , we square and add the first and second equations from 3.1:

$$\begin{aligned} (x \cos \theta_1 - l_1 + y \sin \theta_1)^2 + z^2 &= (l_4 \cos(\theta_3 - \theta_2 + \theta_4) + l_2 \cos \theta_2 + l_3 \cos(\theta_2 - \theta_3))^2 \\ &+ (l_2 \sin \theta_2 - l_4 \sin(\theta_3 - \theta_2 + \theta_4) + l_3 \sin(\theta_2 - \theta_3))^2 \end{aligned}$$

We can make the following substitutions in the above equation:

$$C = (x \cos \theta_1 - l_1 + y \sin \theta_1)^2 + z^2,$$

$$A = \theta_3 - \theta_2 + \theta_4,$$

$$B = \theta_2 - \theta_3.$$

Our equation becomes:

$$C = (l_4 \cos A + l_2 \cos \theta_2 + l_3 \cos B)^2 + (l_2 \sin \theta_2 - l_4 \sin A + l_3 \sin B)^2$$

which is then simplified to:

$$C = l_4^2 + l_2^2 + l_3^2 + 2l_4l_2 \cos(A + \theta_2) + 2l_3l_4 \cos(A + B) + 2l_2l_3 \cos(\theta_2 - B)$$

Substituting back values for A and B :

$$C = l_4^2 + l_2^2 + l_3^2 + 2l_4l_2 \cos(\theta_3 + \theta_4) + 2l_3l_4 \cos \theta_4 + 2l_2l_3 \cos \theta_3$$

$$\frac{C - l_4^2 - l_2^2 - l_3^2 - 2l_2l_3 \cos \theta_3}{2l_4} = l_2 \cos(\theta_3 + \theta_4) + l_3 \cos \theta_4$$

$$\frac{C - l_4^2 - l_2^2 - l_3^2 - 2l_2l_3 \cos \theta_3}{2l_4} = l_2 \cos \theta_3 \cos \theta_4 - l_2 \sin \theta_3 \sin \theta_4 + l_3 \cos \theta_4$$

We can assume the left side to be equal to D :

$$D = \cos \theta_4(l_2 \cos \theta_3 + l_3) - \sin \theta_4(l_2 \sin \theta_3)$$

To simplify the equation, we can make another substitution here:

$$E = l_2 \cos \theta_3 + l_3,$$

$$F = l_2 \sin \theta_3.$$

Thus, we have:

$$D = E \cos \theta_4 - F \sin \theta_4$$

To find a closed-form expression, we can substitute:

$$E = a \sin \phi,$$

$$F = a \cos \phi.$$

Thus, D becomes:

$$D = a \sin \phi \cos \theta_4 - a \cos \phi \sin \theta_4$$

We can obtain a closed-form expression in terms of $\sin \theta$:

$$D = a \sin(\phi - \theta_4)$$

$$\sin^{-1}\left(\frac{D}{a}\right) = \phi - \theta_4$$

$$\boxed{\theta_4 = \phi - \sin^{-1}\left(\frac{D}{a}\right)} \quad (3.3)$$

The expressions for a and ϕ can be determined by dividing the equations for E and F :

$$\frac{E}{F} = \tan \phi \Rightarrow \phi = \tan^{-1}\left(\frac{E}{F}\right)$$

$$a = \frac{E}{\sin \phi}$$

We can also determine θ_4 in terms of $\cos$, which flips the substitutions for E and F previously:

$$E = a \cos \phi,$$

$$F = a \sin \phi.$$

Thus, D becomes:

$$D = a \cos \phi \cos \theta_4 - a \sin \phi \sin \theta_4$$

We can obtain a closed-form expression:

$$D = a \cos(\phi + \theta_4)$$

$$\cos^{-1}\left(\frac{D}{a}\right) = \phi + \theta_4$$

We would have a possibility of plus or minus for $\cos^{-1}$ since it's an even function.

$$\boxed{\theta_4 = -\phi \pm \cos^{-1}\left(\frac{D}{a}\right)}$$

$$\frac{F}{E} = \tan \phi \Rightarrow \phi = \tan^{-1}\left(\frac{F}{E}\right)$$

$$a = \frac{F}{\sin \phi}$$

After having calculated θ_4 , we can determine θ_2 by solving the second equation from the set of IK equations:

$$z = l_2 \sin \theta_2 - l_4 \sin(\theta_3 - \theta_2 + \theta_4) + l_3 \sin(\theta_2 - \theta_3)$$

$$z = l_2 \sin \theta_2 - l_4 \sin(\theta_3 + \theta_4) \cos \theta_2 + l_4 \cos(\theta_3 + \theta_4) \sin \theta_2 + l_3 \sin \theta_2 \cos \theta_3 - l_3 \cos \theta_2 \sin \theta_3$$

$$z = \sin \theta_2 (l_2 + l_4 \cos(\theta_3 + \theta_4) + l_3 \cos \theta_3) - \cos \theta_2 (l_4 \sin(\theta_3 + \theta_4) + l_3 \sin \theta_3)$$

We can make the following substitutions:

$$\alpha = l_2 + l_4 \cos(\theta_3 + \theta_4) + l_3 \cos \theta_3,$$

$$\beta = l_4 \sin(\theta_3 + \theta_4) + l_3 \sin \theta_3$$

$$z = \alpha \sin \theta_2 - \beta \cos \theta_2$$

Substituting the following in the previous equation:

$$\alpha = b \cos \gamma,$$

$$\beta = b \sin \gamma$$

$$z = b \cos \gamma \sin \theta_2 - b \sin \gamma \cos \theta_2$$

$$z = b \sin(\theta_2 - \gamma)$$

$$\sin^{-1}\left(\frac{z}{b}\right) = \theta_2 - \gamma$$

$$\boxed{\theta_2 = \sin^{-1}\left(\frac{z}{b}\right) + \gamma} \quad (3.4)$$

We can find the values of γ and b as follows:

$$\frac{\beta}{\alpha} = \tan \gamma \Rightarrow \gamma = \tan^{-1}\left(\frac{\beta}{\alpha}\right)$$

$$b = \frac{\beta}{\sin \gamma}$$

We can also find an expression for θ_2 in terms of $\cos$ by making the following

substitutions:

$$\alpha = b \sin \gamma,$$

$$\beta = b \cos \gamma$$

$$z = b \sin \gamma \sin \theta_2 - b \cos \gamma \cos \theta_2$$

$$z = -(b \cos \gamma \cos \theta_2 - b \sin \gamma \sin \theta_2)$$

$$z = -b \cos(\gamma + \theta_2)$$

$$\cos^{-1}\left(\frac{-z}{b}\right) = \theta_2 + \gamma$$

$$\boxed{\theta_2 = \pm \cos^{-1}\left(\frac{-z}{b}\right) - \gamma}$$

The values of b and γ can be found below:

$$\frac{\alpha}{\beta} = \tan \gamma \Rightarrow \gamma = \tan^{-1}\left(\frac{\alpha}{\beta}\right)$$

$$b = \frac{\alpha}{\sin \gamma}$$

We keep θ_3 as a constant value since it is the bend angle of the physical link and does not change during motion.

$$\boxed{\theta_3 = \pi/4}$$

The left side of our robot is symmetric to the right side, except for the servos of the knee rotating in the opposite direction, which can be understood from the CAD model of the robot [91]. Thus, the forward and inverse kinematic models for the left legs (FL & BL) can be derived in a similar manner as the right legs, and we will observe the only

difference in the inverse kinematics model to be the negative sign in front of z , as seen below:

$$\begin{pmatrix} x \cos(\theta_1) - l_1 + y \sin(\theta_1) \\ -z \\ x \sin(\theta_1) - y \cos(\theta_1) \end{pmatrix} = \begin{pmatrix} l_4 \cos(\theta_3 - \theta_2 + \theta_4) + l_2 \cos(\theta_2) + l_3 \cos(\theta_2 - \theta_3) \\ l_2 \sin(\theta_2) - l_4 \sin(\theta_3 - \theta_2 + \theta_4) + l_3 \sin(\theta_2 - \theta_3) \\ 0 \end{pmatrix} \quad (3.5)$$

Solving this in the same manner as above, we get the following equations:

$$\boxed{\theta_1 = \arctan 2(y, x)} \quad (3.6)$$

$$\boxed{\theta_3 = -\pi/4} \quad (3.7)$$

$$C = \frac{(x \cos(\theta_1) + y \sin(\theta_1) - L_1)^2 + z^2 - L_2^2 - L_3^2 - L_4^2 - 2L_2L_3 \cos(\theta_3)}{2L_4} \quad (3.8)$$

$$A_1 = L_2 \cos(\theta_3) + L_3 \quad (3.9)$$

$$B_1 = L_2 \sin(\theta_3) \quad (3.10)$$

$$\phi_1 = \arctan 2(B_1, A_1) \quad (3.11)$$

$$a_1 = \sqrt{A_1^2 + B_1^2} \quad (3.12)$$

$$\boxed{\theta_4 = -\arccos\left(\frac{C}{a_1}\right) - \phi_1} \quad (3.13)$$

$$A_2 = L_2 + L_3 \cos(\theta_3) + L_4 \cos(\theta_3 + \theta_4) \quad (3.14)$$

$$B_2 = L_4 \sin(\theta_3 + \theta_4) + L_3 \sin(\theta_3) \quad (3.15)$$

$$\phi_2 = \arctan 2(A_2, B_2) \quad (3.16)$$

$$a_2 = \sqrt{A_2^2 + B_2^2} \quad (3.17)$$

$$\theta_2 = \arccos\left(\frac{z}{a_2}\right) - \phi_2 \quad (3.18)$$

Having obtained the kinematic model, we can now move onto the gait planning. For this controller, we use a crawl gait. For a quadruped, a crawl gait is one where at most one leg is in swing phase at a given time, giving a wider support polygon under the robot with three legs in stance. Hence, the stance phase is three times longer than the swing phase. The different permutations for the swing legs may result in better or worse stability. Typically, alternating sides (left and right) in the swing order helps keep the robot from tilting too far on one side. A possible schedule is given in table 3.2.

Table 3.2: Crawl Gait Schedule

Leg	$0 \leq t < \frac{N}{4}$	$\frac{N}{4} \leq t < \frac{N}{2}$	$\frac{N}{2} \leq t < \frac{3N}{4}$	$\frac{3N}{4} \leq t < N$
FR	Swing	Stance	Stance	Stance
BL	Stance	Swing	Stance	Stance
BR	Stance	Stance	Swing	Stance
FL	Stance	Stance	Stance	Swing

where N is the total number of discrete points in the trajectory for one complete gait cycle. The legs are ordered as Front Right (FR), Back Left (BL), Back Right (BR), and Front Left (FL).

The high-level idea is to have trajectories for the swing and stance phases for each leg, and then use the gait schedule to determine which trajectory to follow at a given